Software Engineering Teaching Unit

Faculty of Science, University of Kelaniya

SYSTEM REQUIREMENT SPECIFICATION(SRS) DOCUMENT

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for the Smart Tourism Platform, a centralized tourism management system designed to assist local and foreign tourists in planning and managing trips throughout Sri Lanka.

The platform addresses several key challenges identified through surveys and interviews with travelers:

1.Fragmented Tourism Services

Tourists currently use multiple platforms for hotel booking, transportation arrangements, tour guide hiring, and destination discovery. This results in:

- Time-consuming planning
- Inconsistent information
- Difficulty comparing options

2.Lack of Hidden Destination Information

Most tourism websites focus only on popular attractions while lesser-known destinations receive limited exposure.

3.Inefficient Trip Planning

Travelers often struggle to coordinate accommodations, transport, guides, and attractions into a single itinerary.

The Smart Tourism Platform provides:

Hotel Booking - Users can search, compare, and reserve hotels based on location, budget, and ratings.
Transport Booking - Users can arrange transportation services including taxis, private vehicles, and tour buses.
Tourist Guide Services - Certified guides can register and offer services to travelers.
Destination Discovery - Users can explore both famous attractions and hidden gems across Sri Lanka.
Personalized Trip Planning - The system generates itineraries based on user preferences.

This document serves as the foundation for analysis, design, implementation, testing, and maintenance activities.

1.2 Scope

1.2.1 In Scope

Component	Description
Tourist Mobile/Web App	Destination search, booking, trip planning
Hotel Management Module	Hotel registration and booking management
Transport Service Module	Vehicle and driver management
Tour Guide Module	Guide registration and booking
Recommendation Engine	Personalized destination suggestions
Admin Dashboard	User management and analytics
Maps Integration	Route planning and location services
Review & Rating System	Tourist feedback

1.2.2 Out of Scope

Item	Reason
International Tourism Services	MVP targets Sri Lanka only
Airline Ticket Booking	Requires airline integrations
Visa Processing	Outside tourism platform scope
Cryptocurrency Payments	Not required for MVP
Augmented Reality Tours	Future roadmap feature

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
GPS	Global Positioning System
API	Application Programming Interface
OTA	Online Travel Agency
JWT	JSON Web Token
Admin	System Administrator
Tourist	End User

1.4 References

- IEEE 830 SRS Standard
- Sri Lanka Tourism Development Authority Publications
- SENG 31242 Project Guidelines
- Tourism User Survey Results

1.5 Overview

The remainder of this document is organized as follows:

- Section 2 describes the overall system.
- Section 3 presents system analysis and use-case modeling.
- Section 4 defines all functional requirements.
- Section 5 specifies non-functional requirements.
- Section 6 evaluates alternative solutions and feasibility.

2. Overall Description

2.1 Product Perspective

The Smart Tourism Platform is a centralized tourism ecosystem connecting tourists, hotels, transport providers, tourist guides, and administrators.

Major Components

- Tourist Mobile/Web Application
- Hotel Management Portal
- Transport Provider Portal
- Tourist Guide Portal
- Admin Dashboard
- Recommendation Engine

2.2 User Classes and Characteristics

Persona 1: Kavinya — Tourist
Demographics: 29 years old, Software Engineer from Germany, Visiting Sri Lanka for a 10-day vacation
Travel Profile: Travels internationally 2–3 times per year, relies heavily on mobile applications for trip planning, prefers online bookings and digital payments
Goals: Discover popular and hidden tourist destinations, Book hotels and transportation easily, Find trusted tourist guides, Create a well-organized travel itinerary
Pain Points: Tourism information is scattered across multiple websites, Difficulty finding authentic local experiences, Uncertainty about transportation options, Limited knowledge of local service providers
Frequency of Use: Uses the platform daily during her trip.

Persona 2: Nimal Perera — Hotel Owner
Demographics: 45 years old, Owner of a boutique hotel in Ella, 15 years of experience in the hospitality industry
Business Profile: Manages 20 guest rooms, employs 12 staff members, primarily attracts foreign tourists
Tech Profile: Comfortable using smartphones and web applications, Uses online booking platforms regularly
Goals: Increase hotel visibility, attract more tourists, manage reservations efficiently, Reduce dependency on third-party booking platforms

Pain Points: High commission fees from external booking websites, Difficulty reaching international customers directly, Managing bookings from multiple platforms
Frequency of Use: Uses the system multiple times daily.

Persona 3: Tharsan — Transport Provider
Demographics: 38 years old, Owner of a small tourist transport service in Jaffna
Business Profile: Operates a fleet of 8 vehicles, Provides airport transfers and intercity transportation,
Serves both local and international tourists
Tech Profile: Uses smartphones and navigation applications daily, Familiar with online booking systems
Goals: Receive more transport bookings, improve vehicle utilization, manage schedules efficiently, Expand customer reach
Pain Points: Seasonal fluctuations in bookings, Difficulty promoting services to tourists, Managing transportation requests manually
Frequency of Use: Uses the platform throughout the day.

Persona 4: Sutharsan — Licensed Tourist Guide
Demographics: 33 years old**,** Licensed tourist guide based in Trincomalee
Professional Profile: 8 years of guiding experience, specializes in cultural and historical tours, Speaks English, Tamil and Sinhala
Tech Profile: Active social media user**,** uses digital platforms to communicate with tourists
Goals: Increase tour bookings**,** build a strong professional reputation**,** promote lesser-known destinations**,** Manage tour schedules efficiently
Pain Points: Limited exposure to international tourists, Dependence on travel agencies for clients,
Manual booking coordination
Frequency of Use: Uses the platform daily.

Persona 5: Ganeshamoorthy Kajenthiranan — System Administrator
Demographics: 23 years old, Platform Operations Administrator
Role: Oversees platform activities, verifies service providers, Resolves complaints and disputes, Monitors system performance
Tech Profile: Advanced computer skills, Familiar with analytics and reporting tools
Goals: Maintain platform quality, ensure trustworthy services, monitor user activity, Generate operational reports
Pain Points: Managing large numbers of provider registrations, Handling user complaints efficiently, Ensuring data accuracy

Frequency of Use: Uses the platform every day.

2.3 Operating Environment

The Smart Tourism Platform will operate in a cross-platform environment, supporting both web and mobile applications. The system is designed to provide seamless access to tourism services for users in Sri Lanka and international tourists.

System Environment

Component	**Requirement**
Mobile Application	Android 10+ / iOS 15+
Web Application	Latest versions of Chrome, Firefox, Edge, Safari
Backend Server	Spring Boot (Java 17+)
Database Server	MySQL
API Architecture	RESTful APIs (Spring Boot)
Mobile Devices	Smartphones and tablets
Desktop Devices	Windows / macOS / Linux
Network	Stable internet connection (4G/5G/Wi-Fi)
Hosting Environment	Cloud platform (AWS / Azure / Render / VPS)
Maps Integration	Google Maps API

2.4 Design and Implementation Constraints

The following constraints influence the design and implementation of the system:

1. Technology Constraint

The backend is implemented using **Spring Boot**, requiring Java-based development and RESTful API architecture.

2. Cross-Platform Requirement

The system must support both **mobile and web applications**, requiring responsive design and mobile-compatible APIs.

3. Third-Party API Dependency

The system relies on external services such as:

- Google Maps API for location services

- Email/SMS services for notifications
- Payment gateway APIs

4. Security Constraint
All user authentication and authorization must be handled using secure mechanisms such as JWT and password hashing (BCrypt).

5. Performance Constraint
The system should handle multiple concurrent users without significant delay in booking and search operations.

6. Development Time Constraint
The project must be completed within the academic semester timeline.

7. Database Constraint
The system must maintain structured and consistent data for users, bookings, and service providers.

2.5 Assumptions and Dependencies

Assumptions

1. Users have access to smartphones or computers with internet connectivity.
2. Tourists, hotels, transport providers, and guides will register with accurate information.
3. Service providers will actively manage their availability and bookings.
4. Users will use the platform for planning and booking tourism services in Sri Lanka.
5. GPS and location services are available for navigation features.

Dependencies

1. **Spring Boot Backend Framework**
    - Used for building RESTful APIs and business logic.
2. **Database System (MongoDB / MySQL)**
    - Stores user data, bookings, and service information.
3. **Google Maps API**
    - Provides mapping, route planning, and location search.
4. **Authentication Service (JWT)**
    - Handles secure login and session management.

5. **Email/SMS Service**
    - Used for booking confirmations and notifications.
6. **Cloud Hosting Platform**
    - Required for deployment and system availability.

3. System Analysis

3.1 Fact-Gathering Techniques Used

To understand user requirements and system expectations for the Smart Tourism Platform, multiple fact-gathering techniques were applied. These techniques helped identify real-world problems and define system improvements.

3.1.1 Survey Analysis

A structured survey was conducted among local and international tourists to analyze travel behavior, booking preferences, and challenges faced during trip planning in Sri Lanka.

The survey revealed that users:

- Depend on multiple platforms for different services
- Face difficulties in budgeting and planning
- Struggle to discover hidden or lesser-known destinations
- Prefer a single integrated travel solution

This confirms the need for a centralized Smart Tourism Platform.

3.1.2 Competitive Analysis

Existing platforms such as Booking.com, Airbnb, and Tripadvisor were analyzed to evaluate features, usability, and limitations.

Key findings:

- Lack of integration between hotel, transport, and guide services
- Limited personalization in travel recommendations
- Weak support for discovering local attractions

These gaps strongly justify the need for an improved unified system.

3.2 Existing System Analysis (with Pain Points)

The current tourism ecosystem in Sri Lanka is highly fragmented. Users must interact with multiple independent systems to complete travel planning.

3.2.1 Lack of Integration

Users must switch between multiple platforms for hotels, transport, and guides, leading to inefficient planning.

3.2.2 Time-Consuming Process

Travel planning requires manual searching, comparison, and coordination across different services.

3.2.3 Limited Real-Time Availability

Transport and guide availability is often not updated in real-time, causing booking delays.

3.2.4 Communication Barriers

Foreign tourists face language difficulties when communicating with local service providers.

3.2.5 Lack of Personalization

Existing systems do not offer personalized travel recommendations based on budget or preferences.

3.2.6 Trust and Reliability Issues

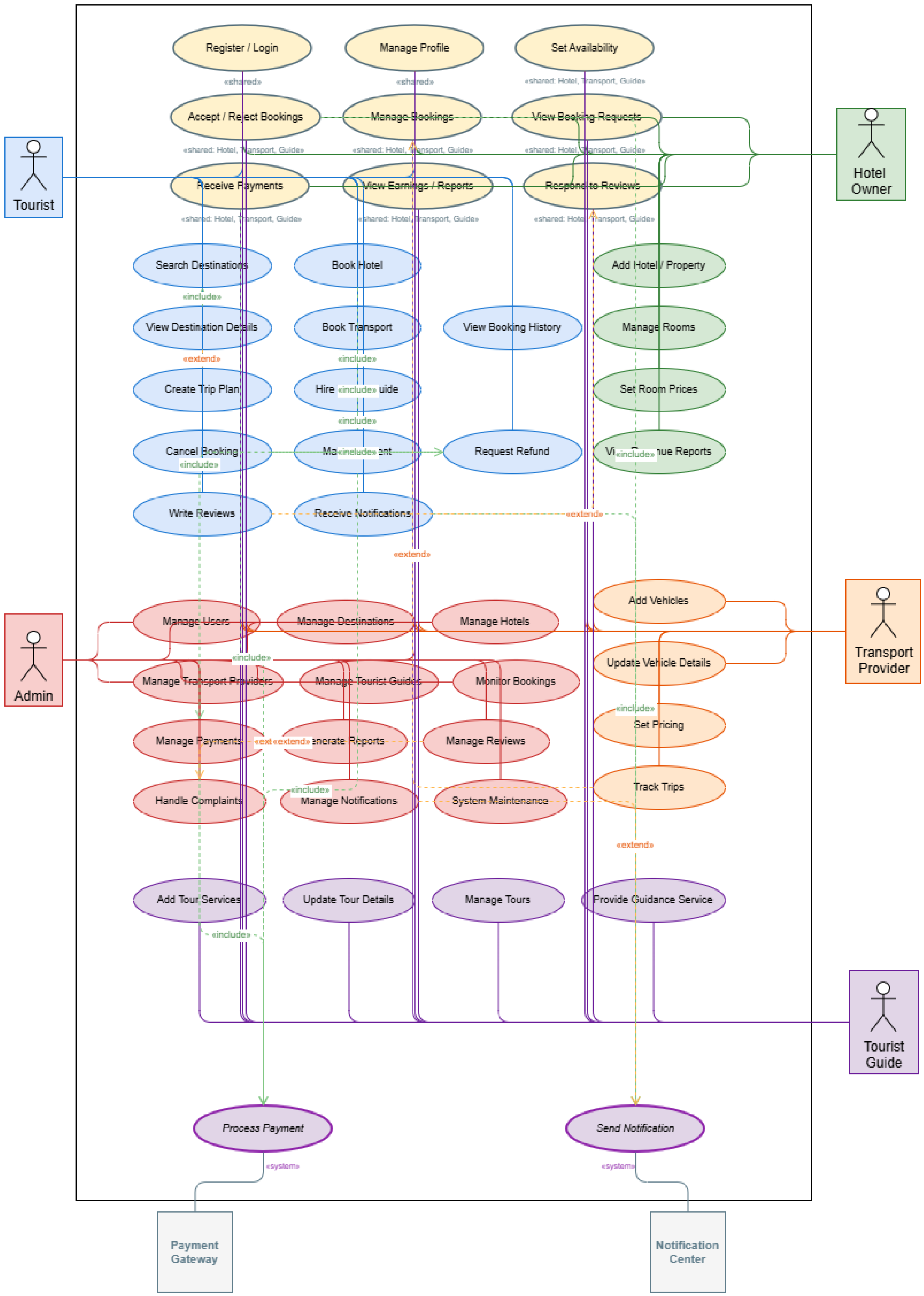
Users find it difficult to verify the authenticity of service providers.

3.2.7 No Unified Mobile Experience

There is no single mobile application that integrates all tourism-related services.

Evidence: **https://github.com/seng31242-travelplanner-2026/research.git**

3.3 Use case diagram



3.4 Use case description

Use Case ID	UC-01
Use Case Name	Register / Login
Actor(s)	Tourist, Hotel Owner, Transport Provider, Tourist Guide (Common to all actors – flows described for Tourist; other actors follow the same pattern with their respective dashboards)
Preconditions	User has internet access and valid credentials for login
Postconditions – Success	User account is created or user is successfully logged into the system
Postconditions – Failure	Registration/login fails; error message displayed
Main Flow	1. User opens the system. 2. Selects Register or Login. 3. Enters required details. 4. System validates information. 5. System creates account or authenticates user. 6. User is redirected to their role dashboard.
Alternative Flow 1	At step 4: If credentials are invalid, system displays error message.
Alternative Flow 2	At step 3: If email already exists during registration, system prompts user to login instead.
Business Rules	BR-01: Email must be unique. BR-02: Password must meet minimum security requirements.

Use Case ID	UC-02
Use Case Name	Manage Profile
Actor(s)	Tourist, Hotel Owner, Transport Provider, Tourist Guide (Common to all actors)
Preconditions	User is logged into the system
Postconditions – Success	Profile information is updated successfully
Postconditions – Failure	Profile remains unchanged; validation error displayed
Main Flow	1. User opens profile settings. 2. System displays current profile information. 3. User edits details. 4. User submits changes.

	5. System validates and saves updates.
Alternative Flow 1	At step 5: If required fields are empty or invalid, system displays validation errors.
Alternative Flow 2	User cancels editing; system returns to dashboard without changes.
Business Rules	BR-03: Contact details (phone, email) must follow valid format requirements.

Use Case ID	UC-03
Use Case Name	Set Availability
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider is logged in and has registered services/properties/vehicles
Postconditions – Success	Availability schedule is updated successfully
Postconditions – Failure	Availability update fails
Main Flow	1. Provider opens availability settings. 2. Selects service, vehicle, or room. 3. Chooses available dates and time slots. 4. System validates information. 5. Availability schedule is saved.
Alternative Flow 1	At step 4: If selected slots overlap or exceed capacity, system displays validation error.
Alternative Flow 2	Provider cancels changes before saving.
Business Rules	BR-04: Resources cannot be assigned to overlapping reservations. Availability cannot exceed total capacity.

Use Case ID	UC-04
Use Case Name	Accept / Reject Bookings
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Pending booking requests exist in the system
Postconditions – Success	Booking request status is updated

Postconditions – Failure	Booking request remains pending
Main Flow	1. Provider views booking requests. 2. Selects a request. 3. Reviews booking details. 4. Accepts or rejects the booking. 5. System updates booking status and notifies the tourist.
Alternative Flow 1	At step 3: If the resource (room/vehicle/guide slot) becomes unavailable, system automatically rejects the request.
Alternative Flow 2	Provider delays decision; request remains pending.
Business Rules	BR-05: Booking decisions must be recorded with timestamps.

Use Case ID	UC-05
Use Case Name	Manage Bookings
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider is logged into the system
Postconditions – Success	Booking details are updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Provider opens booking management section. 2. System displays current bookings. 3. Provider updates booking details or status. 4. System validates changes. 5. Updated booking information is saved.
Alternative Flow 1	At step 4: If update conflicts with another booking, system rejects changes.
Alternative Flow 2	Provider cancels modifications before saving.
Business Rules	BR-06: Double booking of resources is prohibited.

Use Case ID	UC-06
Use Case Name	View Booking Requests

Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Pending booking requests are available in the system
Postconditions – Success	Booking requests are displayed
Postconditions – Failure	Error message displayed
Main Flow	1. Provider selects booking requests option. 2. System retrieves pending requests. 3. System displays request details.
Alternative Flow 1	At step 2: If no requests exist, system displays empty request list.
Alternative Flow 2	Provider filters requests by date, room type, vehicle type, or destination.
Business Rules	BR-07: Requests must be displayed in chronological order.

Use Case ID	UC-07
Use Case Name	Receive Payments
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide, Payment Gateway (Common to service providers)
Preconditions	Tourist payment has been completed
Postconditions – Success	Payment is credited to provider account successfully
Postconditions – Failure	Payment transaction fails
Main Flow	1. Tourist completes payment. 2. Payment gateway processes transaction. 3. System verifies and confirms payment. 4. Provider receives payment notification.
Alternative Flow 1	At step 2: If payment fails, system displays transaction error.
Alternative Flow 2	Payment processing is delayed; system marks transaction as pending.
Business Rules	BR-08: All payment transactions must be securely processed.

Use Case ID	UC-08

Use Case Name	View Earnings / Reports
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Provider has completed bookings with associated transactions
Postconditions – Success	Earnings reports are generated and displayed successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Provider opens reports section. 2. Selects report period. 3. System retrieves transaction data. 4. System generates report. 5. Earnings report is displayed.
Alternative Flow 1	At step 3: If no data exists for the selected period, system displays empty report message.
Alternative Flow 2	Provider exports report for offline use.
Business Rules	BR-09: Reports must include only completed/confirmed transactions.

Use Case ID	UC-09
Use Case Name	Respond to Reviews
Actor(s)	Hotel Owner, Transport Provider, Tourist Guide (Common to service providers)
Preconditions	Tourists have submitted reviews for completed bookings
Postconditions – Success	Response is published successfully
Postconditions – Failure	Response submission fails
Main Flow	1. Provider views customer reviews. 2. Selects a review. 3. Writes response. 4. Submits response. 5. System publishes response.
Alternative Flow 1	At step 4: If response violates content policies, system rejects it.
Alternative Flow 2	Provider edits response before submission.
Business Rules	BR-10: Responses must comply with platform review guidelines and content policies.

Use Case ID	UC-10
Use Case Name	Search Destinations
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Matching destinations are displayed
Postconditions – Failure	No destinations found message displayed
Main Flow	1. Tourist enters search criteria. 2. System processes the request. 3. System retrieves matching destinations. 4. Results are displayed to the tourist.
Alternative Flow 1	At step 3: If no matches are found, system displays "No destinations available".
Alternative Flow 2	Tourist clears filters and performs a new search.
Business Rules	BR-11: Search results must match selected filters and keywords.

Use Case ID	UC-11
Use Case Name	View Destination Details
Actor(s)	Tourist
Preconditions	Tourist has searched for destinations
Postconditions – Success	Destination details are displayed
Postconditions – Failure	Error message displayed if destination unavailable
Main Flow	1. Tourist selects a destination. 2. System retrieves destination details. 3. System displays description, images, pricing, and reviews.
Alternative Flow 1	At step 2: If destination no longer exists, system displays unavailable message.
Alternative Flow 2	Tourist returns to search results page.
Business Rules	BR-12: Only approved destinations are visible to tourists.

Use Case ID	UC-12

Use Case Name	Create Trip Plan
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Trip plan is saved successfully
Postconditions – Failure	Trip plan is not saved; error displayed
Main Flow	1. Tourist selects destinations and services. 2. Tourist enters travel dates and preferences. 3. System generates itinerary. 4. Tourist confirms trip plan. 5. System saves trip plan.
Alternative Flow 1	At step 3: If selected services are unavailable, system suggests alternatives.
Alternative Flow 2	Tourist edits itinerary before confirmation.
Business Rules	BR-13: Travel dates must be valid future dates.

Use Case ID	UC-13
Use Case Name	Book Hotel
Actor(s)	Tourist, Hotel Owner
Preconditions	Tourist is logged in and hotel rooms are available
Postconditions – Success	Hotel booking is confirmed
Postconditions – Failure	Booking is rejected or unavailable
Main Flow	1. Tourist selects hotel. 2. Tourist chooses room and dates. 3. System checks availability. 4. Tourist confirms booking. 5. System records booking and sends confirmation.
Alternative Flow 1	At step 3: If rooms are unavailable, system displays alternative options.
Alternative Flow 2	Tourist cancels booking before confirmation.
Business Rules	BR-14: Double booking of rooms is not allowed.

Use Case ID	UC-14

Use Case Name	Book Transport
Actor(s)	Tourist, Transport Provider
Preconditions	Tourist is logged in and transport is available
Postconditions – Success	Transport booking is confirmed
Postconditions – Failure	Booking fails due to unavailability
Main Flow	1. Tourist selects transport service. 2. Tourist enters travel details. 3. System checks availability. 4. Tourist confirms booking. 5. System records booking and sends notification.
Alternative Flow 1	At step 3: If no vehicles are available, system displays message.
Alternative Flow 2	Tourist modifies travel details and retries booking.
Business Rules	BR-15: Booking must match provider availability schedule.

Use Case ID	UC-15
Use Case Name	Hire Tourist Guide
Actor(s)	Tourist, Tourist Guide
Preconditions	Tourist guide is available
Postconditions – Success	Guide booking is confirmed
Postconditions – Failure	Guide booking request is rejected
Main Flow	1. Tourist browses available guides. 2. Tourist selects a guide. 3. Tourist submits booking request. 4. Guide accepts request. 5. System confirms booking.
Alternative Flow 1	At step 4: If guide rejects request, system suggests other guides.
Alternative Flow 2	Tourist cancels request before acceptance.
Business Rules	BR-16: Guides can only accept bookings during available time slots.

Use Case ID	UC-16
Use Case Name	Make Payment

Actor(s)	Tourist, Payment Gateway
Preconditions	Tourist has an active booking
Postconditions – Success	Payment is processed successfully
Postconditions – Failure	Payment transaction fails
Main Flow	1. Tourist selects payment option. 2. Tourist enters payment details. 3. System forwards request to payment gateway. 4. Gateway processes payment. 5. System confirms successful payment.
Alternative Flow 1	At step 4: If payment is declined, system displays failure message.
Alternative Flow 2	Tourist cancels payment before submission.
Business Rules	BR-17: Payments must use secure encrypted transactions.

Use Case ID	UC-17
Use Case Name	Cancel Booking
Actor(s)	Tourist
Preconditions	Tourist has an active booking
Postconditions – Success	Booking is cancelled successfully
Postconditions – Failure	Booking remains active
Main Flow	1. Tourist views booking history. 2. Tourist selects booking to cancel. 3. System checks cancellation policy. 4. Tourist confirms cancellation. 5. System cancels booking and updates status.
Alternative Flow 1	At step 3: If cancellation deadline passed, system denies cancellation.
Alternative Flow 2	Refund request is generated automatically if applicable.
Business Rules	BR-18: Cancellation policy depends on provider rules.

Use Case ID	UC-18

Use Case Name	View Booking History
Actor(s)	Tourist
Preconditions	Tourist is logged into the system
Postconditions – Success	Booking history is displayed
Postconditions – Failure	Error message displayed
Main Flow	1. Tourist selects booking history option. 2. System retrieves past and current bookings. 3. System displays booking records.
Alternative Flow 1	At step 2: If no bookings exist, system displays empty history message.
Alternative Flow 2	Tourist filters booking history by date or type.
Business Rules	BR-19: Booking records must be stored for audit purposes.

Use Case ID	UC-19
Use Case Name	Write Reviews
Actor(s)	Tourist
Preconditions	Tourist has completed a booking
Postconditions – Success	Review is published successfully
Postconditions – Failure	Review submission fails
Main Flow	1. Tourist selects completed booking. 2. Tourist writes review and rating. 3. Tourist submits review. 4. System validates and publishes review.
Alternative Flow 1	At step 4: If review contains prohibited content, system rejects submission.
Alternative Flow 2	Tourist edits review before final submission.
Business Rules	BR-20: Only verified tourists can submit reviews.

Use Case ID	UC-20
Use Case Name	Receive Notifications
Actor(s)	Tourist, Notification Center
Preconditions	Tourist has enabled notifications

Postconditions – Success	Notification is delivered successfully
Postconditions – Failure	Notification delivery fails
Main Flow	1. System generates notification. 2. Notification center sends message. 3. Tourist receives notification.
Alternative Flow 1	At step 2: If delivery fails, system retries sending notification.
Alternative Flow 2	Tourist disables notifications in settings.
Business Rules	BR-21: Notifications must be sent for booking updates and payments.

Use Case ID	UC-21
Use Case Name	Request Refund
Actor(s)	Tourist, Payment Gateway
Preconditions	Tourist has cancelled a booking eligible for refund
Postconditions – Success	Refund request is submitted and processed
Postconditions – Failure	Refund request is rejected
Main Flow	1. Tourist selects cancelled booking. 2. Tourist requests refund. 3. System verifies eligibility. 4. Refund request is forwarded to payment gateway. 5. System updates refund status.
Alternative Flow 1	At step 3: If booking is not eligible, system displays rejection message.
Alternative Flow 2	Refund processing is delayed; system notifies tourist.
Business Rules	BR-22: Refund amount depends on cancellation policy.

Use Case ID	UC-22
Use Case Name	Add Hotel / Property
Actor(s)	Hotel Owner
Preconditions	Hotel owner is logged into the system
Postconditions – Success	Hotel/property is added successfully

Postconditions – Failure	Property is not added
Main Flow	1. Hotel owner selects add property option. 2. Enters property details. 3. Uploads images and facilities. 4. Submits property information. 5. System validates and saves property details.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Hotel owner cancels property registration before submission.
Business Rules	BR-23: Each property must have a unique registration ID.

Use Case ID	UC-23
Use Case Name	Manage Rooms
Actor(s)	Hotel Owner
Preconditions	Property exists in the system
Postconditions – Success	Room information is updated successfully
Postconditions – Failure	Room information remains unchanged
Main Flow	1. Hotel owner selects property. 2. Opens room management section. 3. Adds, edits, or removes room details. 4. System validates information. 5. Room data is updated.
Alternative Flow 1	At step 4: If room capacity is invalid, system displays error.
Alternative Flow 2	Hotel owner cancels changes before saving.
Business Rules	BR-24: Room details must include capacity and pricing information.

Use Case ID	UC-24
Use Case Name	Set Room Prices
Actor(s)	Hotel Owner
Preconditions	Rooms are registered in the system
Postconditions – Success	Room prices are updated successfully

Postconditions – Failure	Pricing update fails
Main Flow	1. Hotel owner selects room type. 2. Enters pricing details. 3. System validates pricing information. 4. Updated prices are saved.
Alternative Flow 1	At step 3: If price value is invalid, system displays error message.
Alternative Flow 2	Hotel owner schedules seasonal pricing updates.
Business Rules	BR-25: Room prices must be greater than zero.

Use Case ID	UC-25
Use Case Name	View Revenue Reports
Actor(s)	Hotel Owner
Preconditions	Hotel owner has completed financial transactions
Postconditions – Success	Revenue reports are generated successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Hotel owner selects revenue report option. 2. Chooses reporting period. 3. System retrieves transaction records. 4. System generates revenue report. 5. Report is displayed.
Alternative Flow 1	At step 3: If no revenue data exists, system displays empty report message.
Alternative Flow 2	Hotel owner exports report for offline use.
Business Rules	BR-26: Revenue calculations must include only completed transactions.

Use Case ID	UC-26
Use Case Name	Add Vehicles
Actor(s)	Transport Provider
Preconditions	Provider is logged into the system
Postconditions – Success	Vehicle is added successfully

Postconditions – Failure	Vehicle registration fails
Main Flow	1. Provider selects add vehicle option. 2. Enters vehicle details. 3. Uploads required documents. 4. Submits vehicle information. 5. System validates and saves vehicle details.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Provider cancels vehicle registration before submission.
Business Rules	BR-27: Each vehicle must have a unique registration number.

Use Case ID	UC-27
Use Case Name	Update Vehicle Details
Actor(s)	Transport Provider
Preconditions	Vehicle exists in the system
Postconditions – Success	Vehicle details are updated successfully
Postconditions – Failure	Vehicle details remain unchanged
Main Flow	1. Provider selects vehicle. 2. Opens edit vehicle details option. 3. Updates vehicle information. 4. System validates changes. 5. Updated details are saved successfully.
Alternative Flow 1	At step 4: If invalid data is entered, system displays error message.
Alternative Flow 2	Provider cancels modifications before saving.
Business Rules	BR-28: Vehicle capacity and license information must be valid.

Use Case ID	UC-28
Use Case Name	Set Pricing
Actor(s)	Transport Provider
Preconditions	Vehicles are registered in the system
Postconditions – Success	Pricing details are updated successfully

Postconditions – Failure	Pricing update fails
Main Flow	1. Provider selects vehicle or service type. 2. Enters pricing details. 3. System validates price information. 4. Updated pricing is saved successfully.
Alternative Flow 1	At step 3: If price is invalid, system displays error message.
Alternative Flow 2	Provider schedules seasonal price updates.
Business Rules	BR-29: Service price must be greater than zero.

Use Case ID	UC-29
Use Case Name	Track Trips
Actor(s)	Transport Provider
Preconditions	Active trips exist in the system
Postconditions – Success	Trip status and location are displayed
Postconditions – Failure	Trip tracking information unavailable
Main Flow	1. Provider opens trip tracking section. 2. System retrieves active trip data. 3. Current trip status and route are displayed. 4. Provider monitors ongoing trips.
Alternative Flow 1	At step 2: If GPS/location data is unavailable, system displays tracking error.
Alternative Flow 2	Provider refreshes trip status manually.
Business Rules	BR-30: Trip updates must refresh in real time whenever possible.

Use Case ID	UC-30
Use Case Name	Add Tour Services
Actor(s)	Tourist Guide
Preconditions	Tourist guide is logged into the system
Postconditions – Success	Tour service is added successfully

Postconditions – Failure	Tour service creation fails
Main Flow	1. Tourist guide selects add service option. 2. Enters tour service details. 3. Uploads images and descriptions. 4. Submits service information. 5. System validates and saves the service.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Tourist guide cancels service creation before submission.
Business Rules	BR-31: Each tour service must have a unique identifier.

Use Case ID	UC-31
Use Case Name	Update Tour Details
Actor(s)	Tourist Guide
Preconditions	Tour service exists in the system
Postconditions – Success	Tour details are updated successfully
Postconditions – Failure	Tour details remain unchanged
Main Flow	1. Tourist guide selects existing tour service. 2. Opens edit option. 3. Updates service information. 4. System validates changes. 5. Updated details are saved successfully.
Alternative Flow 1	At step 4: If invalid data is entered, system displays validation errors.
Alternative Flow 2	Tourist guide cancels modifications before saving.
Business Rules	BR-32: Tour pricing and schedule must be valid.

Use Case ID	UC-32
Use Case Name	Manage Tours
Actor(s)	Tourist Guide
Preconditions	Tour services are registered in the system
Postconditions – Success	Tour information is managed successfully

Postconditions – Failure	Changes are not saved
Main Flow	1. Tourist guide opens manage tours section. 2. System displays available tours. 3. Guide adds, edits, or removes tours. 4. System validates updates. 5. Changes are saved successfully.
Alternative Flow 1	At step 4: If scheduling conflicts exist, system rejects changes.
Alternative Flow 2	Tourist guide cancels modifications before saving.
Business Rules	BR-33: Tour schedules must not overlap with confirmed bookings.

Use Case ID	UC-33
Use Case Name	Provide Guidance Service
Actor(s)	Tourist Guide, Tourist
Preconditions	Booking is confirmed and tour date has arrived
Postconditions – Success	Guidance service is completed successfully
Postconditions – Failure	Service is interrupted or cancelled
Main Flow	1. Tourist guide reviews scheduled tour. 2. Meets tourist at designated location. 3. Conducts guided tour activities. 4. Provides destination information and assistance. 5. Marks service as completed in the system.
Alternative Flow 1	At step 2: If tourist does not arrive, system records no-show status.
Alternative Flow 2	Tour is cancelled due to weather or emergency conditions.
Business Rules	BR-34: Guides must provide services only for confirmed bookings.

Use Case ID	UC-34
Use Case Name	Admin Login
Actor(s)	Admin
Preconditions	Admin account exists in the system
Postconditions – Success	Admin is authenticated and dashboard is displayed

Postconditions – Failure	Login fails; error message displayed
Main Flow	1. Admin opens login page. 2. Enters username and password. 3. System validates credentials. 4. System authenticates admin. 5. Admin dashboard is displayed.
Alternative Flow 1	At step 3: If credentials are invalid, system displays error message.
Alternative Flow 2	At step 2: If fields are empty, system requests required information.
Business Rules	BR-35: Only authorized administrators can access admin functions.

Use Case ID	UC-35
Use Case Name	Manage Users
Actor(s)	Admin
Preconditions	Admin is logged into the system
Postconditions – Success	User records are updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens user management section. 2. System displays registered users. 3. Admin searches or selects a user. 4. Admin updates, activates, or deactivates user account. 5. System saves changes.
Alternative Flow 1	At step 3: If user does not exist, system displays error message.
Alternative Flow 2	Admin cancels modifications before saving.
Business Rules	BR-36: Admin can suspend accounts violating platform policies.

Use Case ID	UC-36
Use Case Name	Manage Destinations
Actor(s)	Admin
Preconditions	Admin is authenticated
Postconditions – Success	Destination information is updated successfully

Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens destination management section. 2. Views destination list. 3. Adds, edits, or removes destination details. 4. System validates updates. 5. System saves destination information.
Alternative Flow 1	At step 4: If mandatory fields are missing, system displays validation errors.
Alternative Flow 2	Admin cancels changes before saving.
Business Rules	BR-37: Only approved destinations can be published to tourists.

Use Case ID	UC-37
Use Case Name	Manage Hotels
Actor(s)	Admin
Preconditions	Hotel properties exist in the system
Postconditions – Success	Hotel records are managed successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens hotel management section. 2. System displays hotel records. 3. Admin reviews hotel details. 4. Admin approves, updates, or removes hotel information. 5. System saves updates.
Alternative Flow 1	At step 3: If hotel details are invalid, admin rejects registration.
Alternative Flow 2	Admin cancels action before confirmation.
Business Rules	BR-38: Hotels must meet platform verification requirements before approval.

Use Case ID	UC-38
Use Case Name	Manage Transport Providers
Actor(s)	Admin
Preconditions	Transport providers are registered in the system

Postconditions – Success	Provider information is updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens transport provider management section. 2. System displays provider records. 3. Admin reviews provider details. 4. Admin approves, updates, or removes provider information. 5. System saves updates.
Alternative Flow 1	At step 3: If provider documents are invalid, registration is rejected.
Alternative Flow 2	Admin suspends provider account for policy violations.
Business Rules	BR-39: Providers must submit valid license information.

Use Case ID	UC-39
Use Case Name	Manage Tourist Guides
Actor(s)	Admin
Preconditions	Tourist guides are registered in the system
Postconditions – Success	Guide information is updated successfully
Postconditions – Failure	Changes are not saved
Main Flow	1. Admin opens tourist guide management section. 2. System displays guide records. 3. Admin reviews guide details and certifications. 4. Admin approves, updates, or removes guide profiles. 5. System saves changes.
Alternative Flow 1	At step 3: If certifications are invalid, registration is rejected.
Alternative Flow 2	Admin suspends guide account due to complaints or violations.
Business Rules	BR-40: Tourist guides must provide valid certification documents.

Use Case ID	UC-40
Use Case Name	Monitor Bookings
Actor(s)	Admin
Preconditions	Booking records exist in the system

Postconditions – Success	Booking information is monitored successfully
Postconditions – Failure	Booking data cannot be retrieved
Main Flow	1. Admin opens booking monitoring section. 2. System retrieves booking records. 3. Admin reviews booking activities and statuses. 4. System displays booking analytics.
Alternative Flow 1	At step 2: If records are unavailable, system displays error message.
Alternative Flow 2	Admin filters bookings by provider, tourist, or date.
Business Rules	BR-41: All booking activities must be logged for auditing purposes.

Use Case ID	UC-41
Use Case Name	Manage Payments
Actor(s)	Admin, Payment Gateway
Preconditions	Payment transactions exist in the system
Postconditions – Success	Payment records are updated and verified
Postconditions – Failure	Payment update fails
Main Flow	1. Admin opens payment management section. 2. System retrieves transaction records. 3. Admin reviews payment status. 4. Admin verifies or resolves payment issues. 5. System updates payment records.
Alternative Flow 1	At step 3: If transaction fails, admin marks payment as disputed or pending.
Alternative Flow 2	Admin processes refund requests manually when required.
Business Rules	BR-42: Payment records must be securely stored and traceable.

Use Case ID	UC-42
Use Case Name	Generate Reports
Actor(s)	Admin
Preconditions	System contains operational data

Postconditions – Success	Reports are generated successfully
Postconditions – Failure	Report generation fails
Main Flow	1. Admin selects report generation option. 2. Chooses report type and period. 3. System retrieves required data. 4. System generates report. 5. Report is displayed or exported.
Alternative Flow 1	At step 3: If data is unavailable, system displays empty report message.
Alternative Flow 2	Admin exports report in downloadable format.
Business Rules	BR-43: Reports must contain accurate and up-to-date system data.

Use Case ID	UC-43
Use Case Name	Manage Reviews
Actor(s)	Admin
Preconditions	Reviews exist in the system
Postconditions – Success	Reviews are moderated successfully
Postconditions – Failure	Review moderation action fails
Main Flow	1. Admin opens review management section. 2. System displays submitted reviews. 3. Admin reviews reported or flagged content. 4. Admin approves or removes reviews. 5. System updates review status.
Alternative Flow 1	At step 3: If review violates policies, system removes it automatically after confirmation.
Alternative Flow 2	Admin restores previously removed review after appeal review.
Business Rules	BR-44: Offensive or misleading reviews must be removed.

Use Case ID	UC-44
Use Case Name	Handle Complaints
Actor(s)	Admin
Preconditions	Complaints have been submitted by users

Postconditions – Success	Complaint is resolved and updated
Postconditions – Failure	Complaint remains unresolved
Main Flow	1. Admin opens complaint management section. 2. System displays submitted complaints. 3. Admin reviews complaint details. 4. Admin investigates issue. 5. Admin resolves complaint and updates status.
Alternative Flow 1	At step 4: If additional evidence is needed, admin requests more information from users.
Alternative Flow 2	Complaint is escalated to higher authority for further review.
Business Rules	BR-45: Complaints must be resolved within the defined response period.

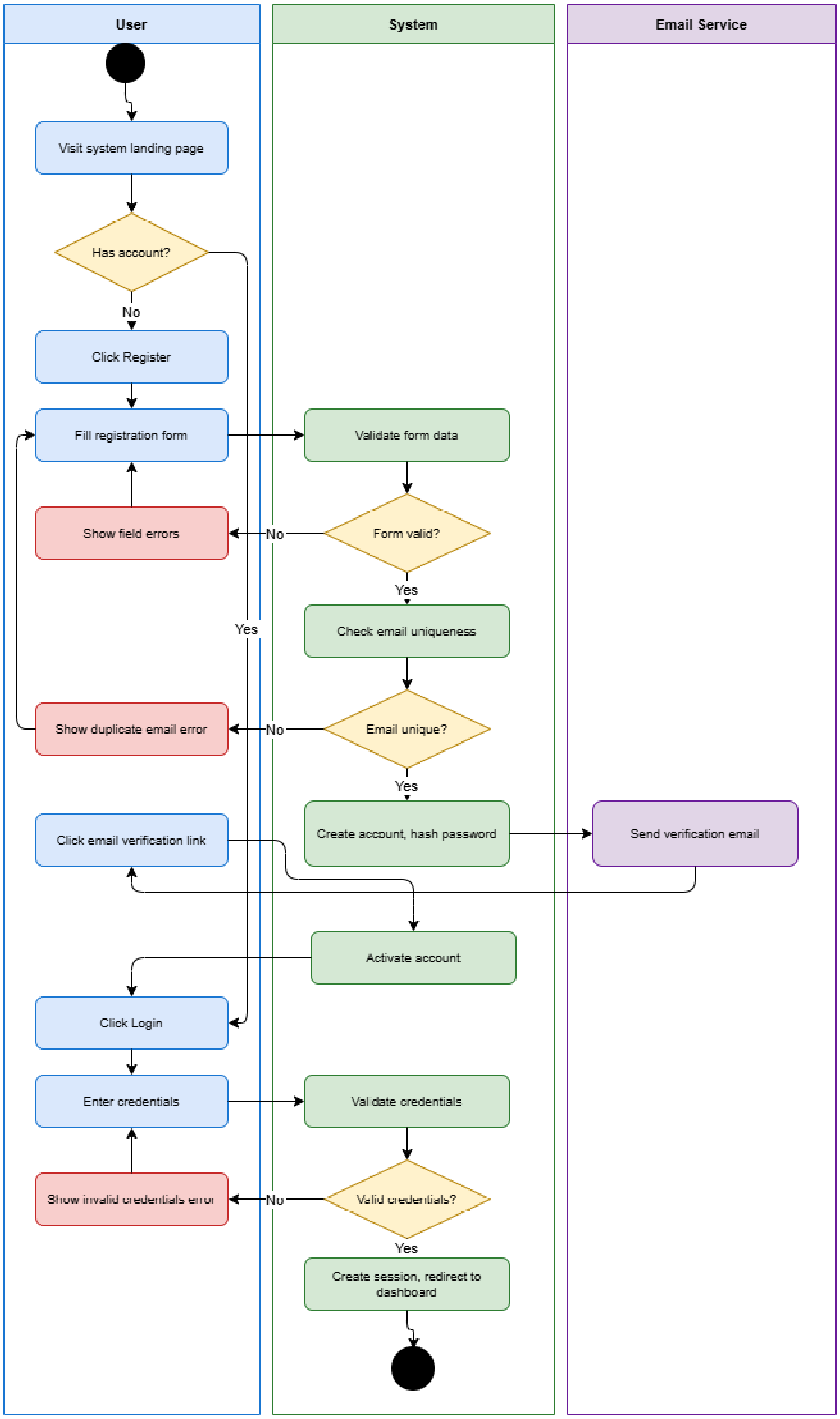
Use Case ID	UC-45
Use Case Name	Manage Notifications
Actor(s)	Admin, Notification Center
Preconditions	Notification service is active
Postconditions – Success	Notifications are created and delivered successfully
Postconditions – Failure	Notification delivery fails
Main Flow	1. Admin opens notification management section. 2. Creates or schedules notification. 3. System validates notification details. 4. Notification center sends messages to users. 5. System logs delivery status.
Alternative Flow 1	At step 4: If delivery fails, system retries notification delivery.
Alternative Flow 2	Admin cancels scheduled notification before sending.
Business Rules	BR-46: Critical notifications must be prioritized for immediate delivery.

Use Case ID	UC-46
Use Case Name	System Maintenance
Actor(s)	Admin

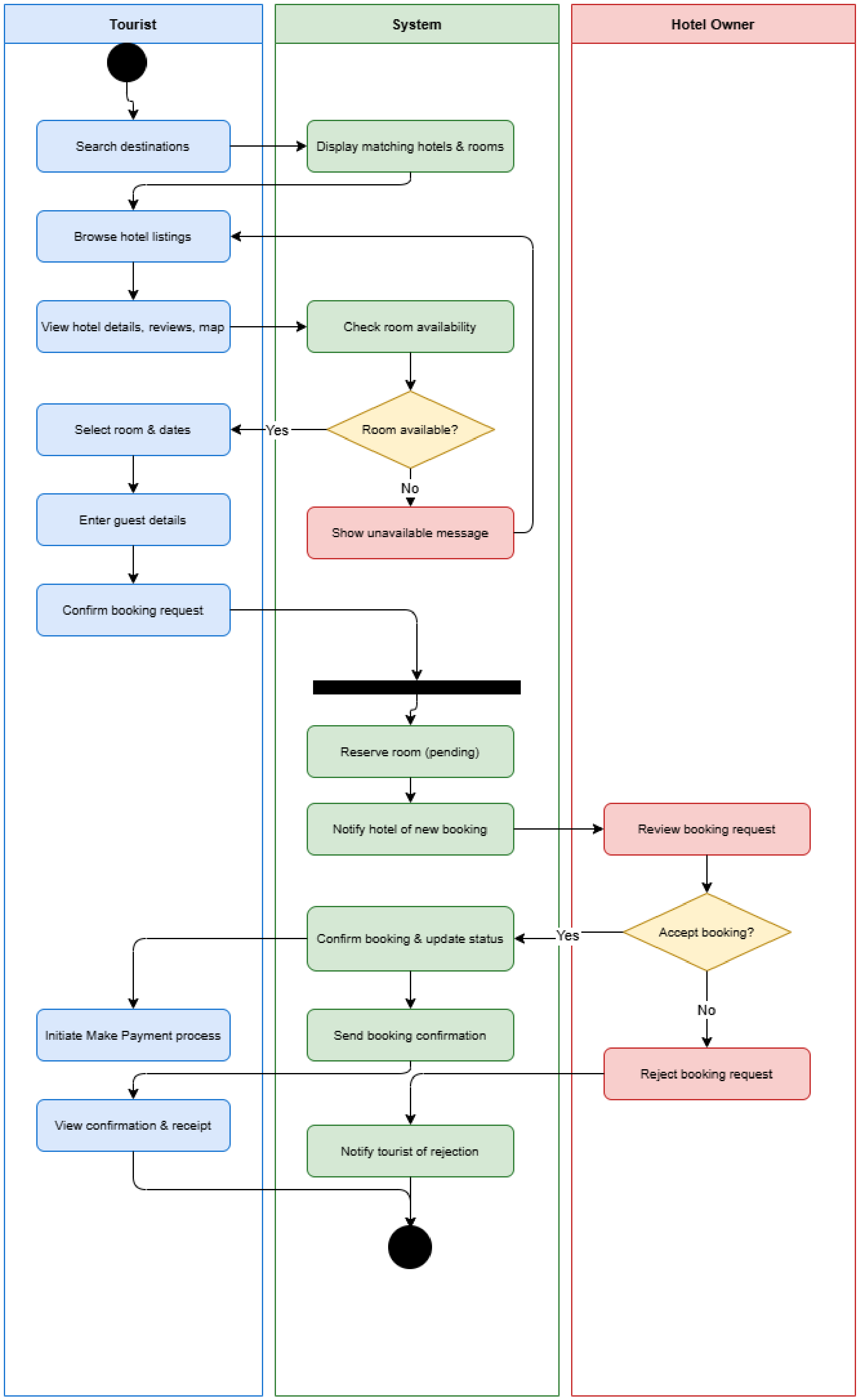
Preconditions	Admin has maintenance privileges
Postconditions – Success	System maintenance tasks are completed successfully
Postconditions – Failure	Maintenance task fails or system remains unstable
Main Flow	1. Admin opens maintenance panel. 2. Selects maintenance task (backup, update, cleanup, etc.). 3. System performs maintenance operation. 4. System displays completion status.
Alternative Flow 1	At step 3: If maintenance operation fails, system logs error details.
Alternative Flow 2	Admin schedules maintenance for a later time.
Business Rules	BR-47: Regular backups must be performed before major system updates.

3.5 Activity diagrams

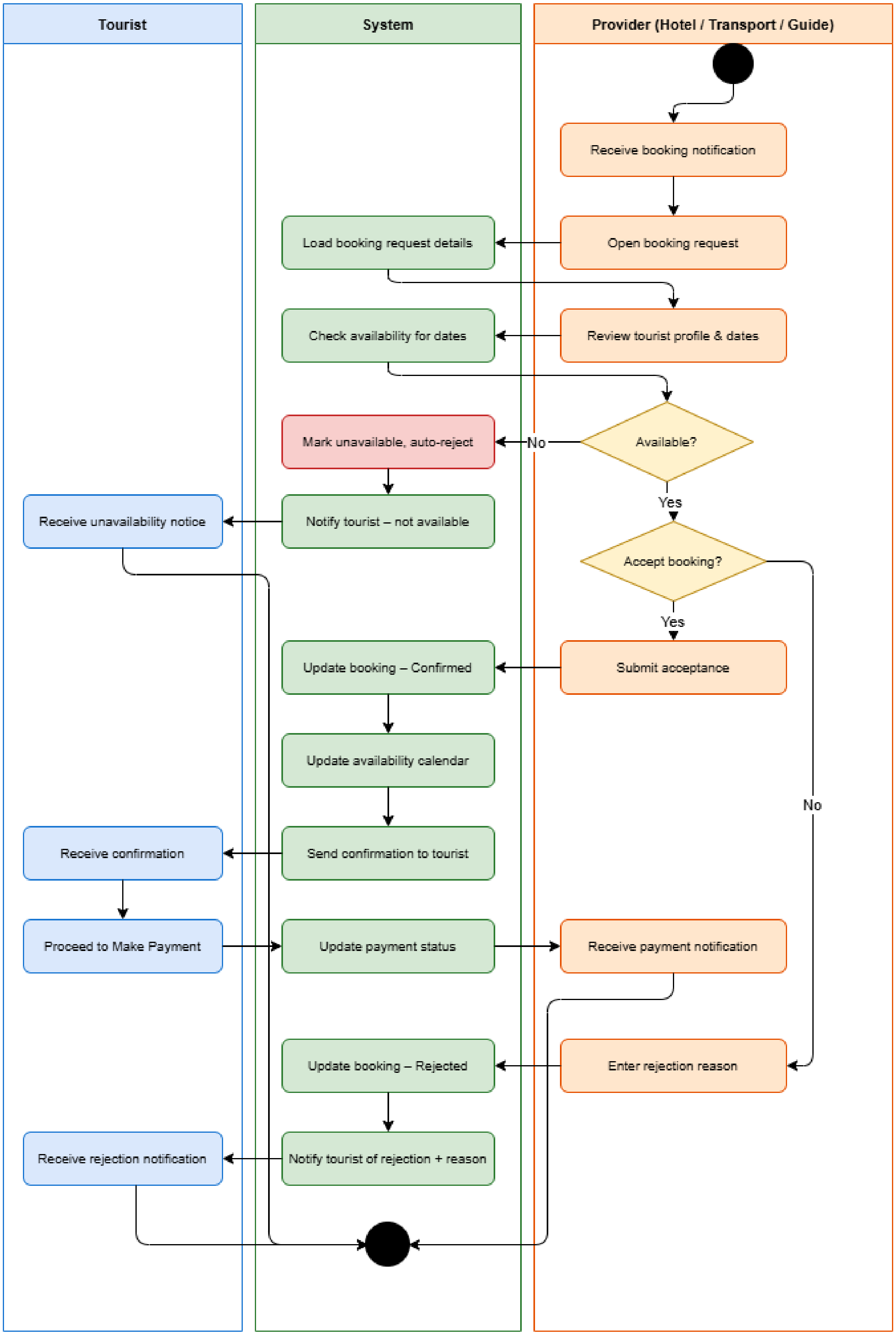
UC-01: Register/Login



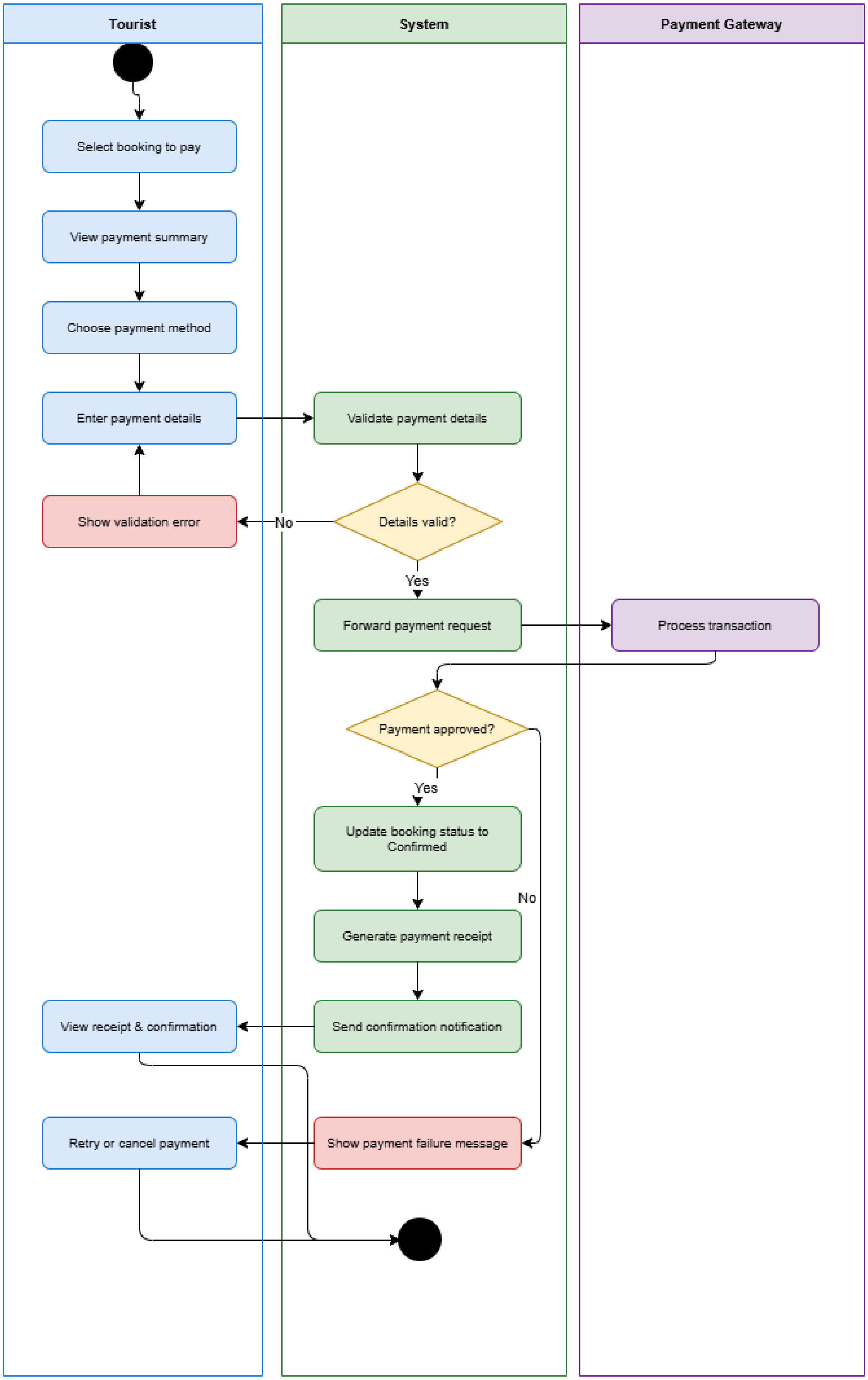
UC-13: Book Hotel



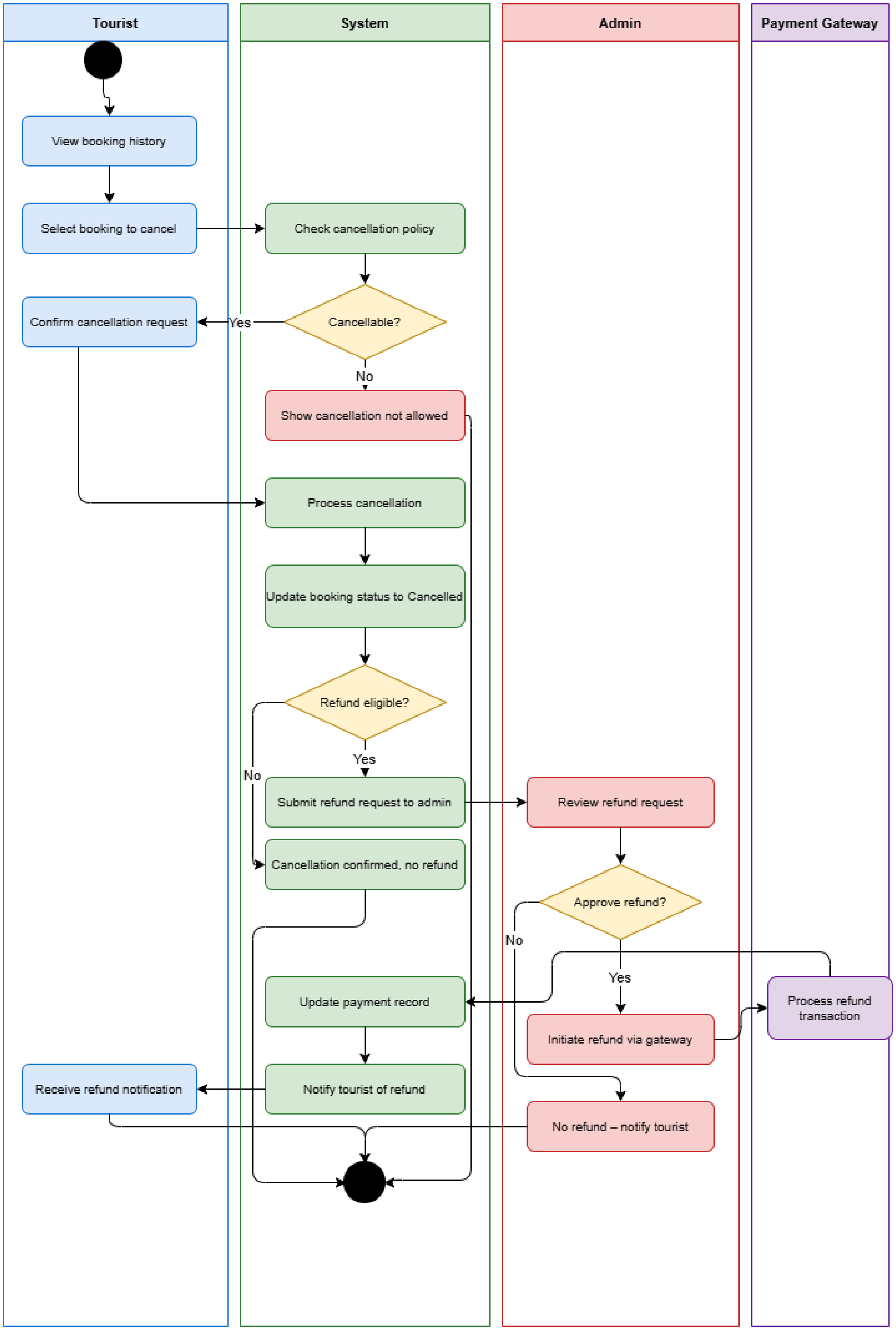
UC-04: Accept/ Reject Bookings



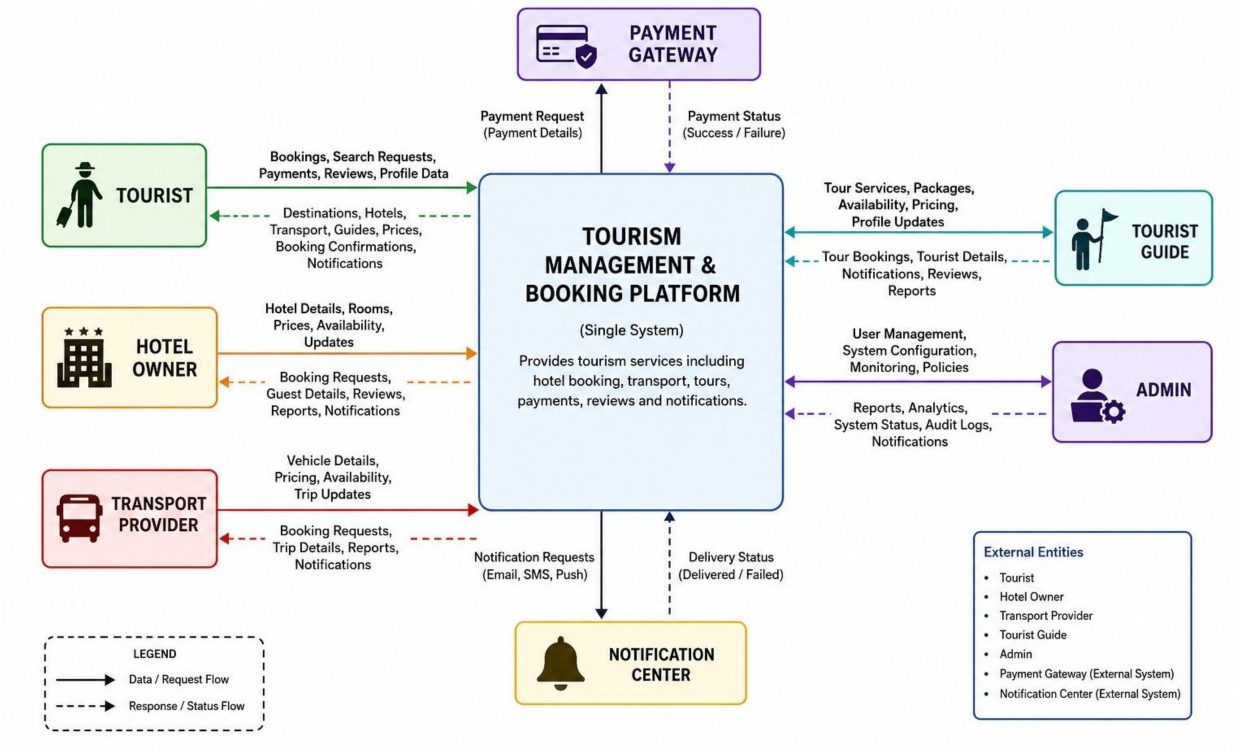
UC-16: Make Payment



UC-17: Cancel Booking & UC-21: Refund



3.6 Context diagrams



4. Functional Requirements

ID	FR-01
Title	User Registration
Description	The system shall allow tourists, hotel owners, transport providers, and tourist guides to register using valid details.
Priority	P1 – Critical
Source	UC-01
Verification	TC-FR-01

ID	FR-02
Title	User Login
Description	The system shall allow registered users to log in securely based on their role.
Priority	P1 – Critical
Source	UC-01
Verification	TC-FR-02

ID	FR-03
Title	Role-Based Dashboard
Description	The system shall redirect users to the correct dashboard based on their role after login.

Priority	P1 – Critical
Source	UC-01
Verification	TC-FR-03

ID	FR-34
Title	Admin Login
Description	The system shall allow administrators to log in securely.
Priority	P1 – Critical
Source	UC-34
Verification	TC-FR-34

ID	FR-05
Title	Set Availability
Description	The system shall allow hotel owners, transport providers, and tourist guides to set their availability schedules.
Priority	P1 – Critical
Source	UC-03
Verification	TC-FR-05

ID	FR-06

Title	View Booking Requests
Description	The system shall allow service providers to view pending booking requests.
Priority	P1 – Critical
Source	UC-06
Verification	TC-FR-06

ID	FR-07
Title	Accept or Reject Bookings
Description	The system shall allow service providers to accept or reject booking requests.
Priority	P1 – Critical
Source	UC-04
Verification	TC-FR-07

ID	FR-08
Title	Manage Bookings
Description	The system shall allow service providers to update booking details and booking statuses.
Priority	P1 – Critical
Source	UC-05

Verification	TC-FR-08

ID	FR-09
Title	Search Destinations
Description	The system shall allow tourists to search tourist destinations using keywords and filters.
Priority	P1 – Critical
Source	UC-10
Verification	TC-FR-09

ID	FR-10
Title	View Destination Details
Description	The system shall display destination details including description, images, pricing, and reviews.
Priority	P1 – Critical
Source	UC-11
Verification	TC-FR-10

ID	FR-12
Title	Book Hotel

Description	The system shall allow tourists to book available hotel rooms.
Priority	P1 – Critical
Source	UC-13
Verification	TC-FR-12

ID	FR-13
Title	Book Transport
Description	The system shall allow tourists to book available transport services.
Priority	P1 – Critical
Source	UC-14
Verification	TC-FR-13

ID	FR-14
Title	Hire Tourist Guide
Description	The system shall allow tourists to browse and hire available tourist guides.
Priority	P1 – Critical
Source	UC-15
Verification	TC-FR-14

ID	FR-15
Title	Make Payment
Description	The system shall allow tourists to make secure online payments for bookings.
Priority	P1 – Critical
Source	UC-16
Verification	TC-FR-15

ID	FR-16
Title	Cancel Booking
Description	The system shall allow tourists to cancel bookings according to provider cancellation policies.
Priority	P1 – Critical
Source	UC-17
Verification	TC-FR-16

ID	FR-48
Title	Transparent Pricing
Description	The system shall display clear pricing details, including booking costs and possible additional charges.
Priority	P1 – Critical

Source	User Persona Report
Verification	TC-FR-48

ID	FR-22
Title	Add Hotel or Property
Description	The system shall allow hotel owners to add hotel or property details with images and facilities.
Priority	P1 – Critical
Source	UC-22
Verification	TC-FR-22

ID	FR-23
Title	Manage Rooms
Description	The system shall allow hotel owners to add, edit, or remove room details.
Priority	P1 – Critical
Source	UC-23
Verification	TC-FR-23

ID	FR-24

Title	Set Room Prices
Description	The system shall allow hotel owners to set and update room pricing.
Priority	P1 – Critical
Source	UC-24
Verification	TC-FR-24

ID	FR-26
Title	Add Vehicles
Description	The system shall allow transport providers to add vehicle details and required documents.
Priority	P1 – Critical
Source	UC-26
Verification	TC-FR-26

ID	FR-27
Title	Update Vehicle Details
Description	The system shall allow transport providers to update existing vehicle information.
Priority	P1 – Critical
Source	UC-27

Verification	TC-FR-27

ID	FR-28
Title	Set Transport Pricing
Description	The system shall allow transport providers to set and update transport service prices.
Priority	P1 – Critical
Source	UC-28
Verification	TC-FR-28

ID	FR-30
Title	Add Tour Services
Description	The system shall allow tourist guides to add tour service details, images, and descriptions.
Priority	P1 – Critical
Source	UC-30
Verification	TC-FR-30

ID	FR-31
Title	Update Tour Details

Description	The system shall allow tourist guides to update tour details, schedule, and pricing.
Priority	P1 – Critical
Source	UC-31
Verification	TC-FR-31

ID	FR-32
Title	Manage Tours
Description	The system shall allow tourist guides to add, edit, or remove tour services.
Priority	P1 – Critical
Source	UC-32
Verification	TC-FR-32

ID	FR-35
Title	Manage Users
Description	The system shall allow administrators to view, activate, deactivate, or update user accounts.
Priority	P1 – Critical
Source	UC-35
Verification	TC-FR-35

ID	FR-41
Title	Manage Payments
Description	The system shall allow administrators to view, verify, and resolve payment transaction issues.
Priority	P1 – Critical
Source	UC-41
Verification	TC-FR-41

ID	FR-17
Title	View Booking History
Description	The system shall allow tourists to view current and past booking records.
Priority	P2 – High
Source	UC-18
Verification	TC-FR-17

ID	FR-21
Title	Receive Notifications
Description	The system shall send booking, payment, approval, and service update notifications to users.

Priority	P2 – High
Source	UC-20, UC-45
Verification	TC-FR-21

ID	FR-52
Title	Mobile-Friendly Access
Description	The system shall provide a mobile-friendly interface for tourists and service providers.
Priority	P2 – High
Source	User Persona Report / Scope
Verification	TC-FR-52

5. Non-Functional Requirements

5.1 Performance

ID	NFR-PE-01
Title	Page Load Time
Description	The system shall load any page within 3 seconds under normal network conditions with up to 500 concurrent users.
Rationale	Slow load times reduce user satisfaction and increase drop-off rates.
Metric / Test	Page load time ≤ 3 seconds (measured via browser performance tools)
Priority	P1 – Critical

ID	NFR-PE-02
Title	Search Response Time
Description	The system shall return destination and service search results within 2 seconds for any query.
Rationale	Tourists expect near-instant search feedback when browsing destinations.
Metric / Test	Search API response time ≤ 2 seconds under load testing
Priority	P1 – Critical

ID	NFR-PE-03
Title	Payment Processing Time
Description	The system shall complete payment transactions within 5 seconds under normal operating conditions.
Rationale	Delayed payment responses cause user anxiety and abandoned transactions.
Metric / Test	Payment gateway round-trip ≤ 5 seconds (99th percentile)
Priority	P1 – Critical

ID	NFR-PE-04
Title	Concurrent User Support
Description	The system shall support a minimum of 1,000 concurrent users without performance degradation.
Rationale	Peak tourist seasons demand high simultaneous usage of the platform.
Metric / Test	Response time remains within SLA thresholds during load tests at 1,000 concurrent users
Priority	P2 – High

5.2 Security

ID	NFR-SE-01
Title	Data Encryption

Description	The system shall encrypt all data in transit using TLS 1.2 or higher and all sensitive data at rest using AES-256.
Rationale	User personal and payment data must be protected from interception and unauthorized access.
Metric / Test	TLS version verified via SSL audit; AES-256 confirmed in database configuration
Priority	P1 – Critical

ID	NFR-SE-02
Title	Authentication Security
Description	The system shall enforce strong password policies and support multi-factor authentication (MFA) for all user roles.
Rationale	Credential theft is a leading cause of unauthorized account access.
Metric / Test	Password policy enforced at registration; MFA available and tested for all roles
Priority	P1 – Critical

ID	NFR-SE-03
Title	Session Management
Description	The system shall automatically expire inactive user sessions after 30 minutes and invalidate tokens on logout.
Rationale	Stale sessions expose accounts to unauthorized access on shared devices.

Metric / Test	Session timeout triggered at 30 minutes of inactivity confirmed via testing
Priority	P1 – Critical

ID	NFR-SE-04
Title	Role-Based Access Control
Description	The system shall restrict access to features and data strictly according to the authenticated user's role.
Rationale	Unauthorized access to admin or provider functions poses operational and legal risk.
Metric / Test	Access control verified for all roles via penetration testing and test cases
Priority	P1 – Critical

ID	NFR-SE-05
Title	SQL Injection & XSS Prevention
Description	The system shall validate and sanitize all user inputs to prevent SQL injection, cross-site scripting (XSS), and other injection attacks.
Rationale	Input-based attacks are among the most common and damaging web vulnerabilities.
Metric / Test	Zero critical vulnerabilities reported in OWASP Top 10 security audit
Priority	P1 – Critical

ID	NFR-SE-06
Title	Payment Data Security
Description	The system shall comply with PCI-DSS standards for all payment data handling and shall not store raw card details.
Rationale	Regulatory compliance is mandatory for platforms processing online payments.
Metric / Test	PCI-DSS compliance confirmed; no raw card data stored in database
Priority	P1 – Critical

5.3 Usability

ID	NFR-US-01
Title	Intuitive Navigation
Description	The system shall enable new users to complete core tasks (registration, search, booking) without external guidance within 5 minutes.
Rationale	Tourist users may be first-time visitors unfamiliar with the platform layout.
Metric / Test	Usability testing with 5 new users; all complete core tasks within 5 minutes
Priority	P2 – High

ID	NFR-US-02
Title	Responsive Design

Description	The system shall provide a fully functional and visually consistent interface across desktop, tablet, and mobile screen sizes.
Rationale	Tourists frequently use mobile devices while traveling to access bookings and services.
Metric / Test	UI tested on Chrome, Safari, Firefox across screen widths 375px–1920px
Priority	P1 – Critical

ID	NFR-US-03
Title	Error Messaging
Description	The system shall display clear, user-friendly error messages that guide users to correct their input or resolve issues.
Rationale	Vague error messages frustrate users and increase support requests.
Metric / Test	All error states reviewed; no generic error codes exposed to end users
Priority	P2 – High

ID	NFR-US-04
Title	Accessibility
Description	The system shall conform to WCAG 2.1 Level AA accessibility standards to support users with disabilities.
Rationale	Inclusive design ensures the platform is accessible to all tourists regardless of ability.

Metric / Test	Accessibility audit using automated tool (axe/Lighthouse) with zero Level AA violations
Priority	P2 – High

5.4 Reliability

ID	NFR-RE-01
Title	System Uptime
Description	The system shall maintain a minimum uptime of 99.5% measured monthly, excluding scheduled maintenance windows.
Rationale	Tourists book services at any time; downtime leads to lost revenue and trust.
Metric / Test	Monthly uptime ≥ 99.5% tracked via uptime monitoring service
Priority	P1 – Critical

ID	NFR-RE-02
Title	Fault Tolerance
Description	The system shall continue to function with degraded but operational service in the event of a single component failure.
Rationale	Single points of failure can cause complete service outages affecting all users.
Metric / Test	Component failure simulation tests confirm graceful degradation with no full outage

Priority	P1 – Critical

ID	NFR-RE-03
Title	Data Backup
Description	The system shall perform automated daily backups of all data and support restoration within 4 hours of a failure event.
Rationale	Data loss is unacceptable for a booking platform with financial transaction records.
Metric / Test	Backup logs verified daily; restoration drill completed within 4-hour RTO
Priority	P1 – Critical

ID	NFR-RE-04
Title	Error Recovery
Description	The system shall automatically recover from transient errors (e.g., database timeouts, API failures) using retry logic without user intervention.
Rationale	Transient failures should not surface to users as hard errors when retry is possible.
Metric / Test	Retry logic unit tested; transient errors resolved transparently in integration tests
Priority	P2 – High

5.5 Scalability

ID	NFR-SC-01
Title	Horizontal Scalability
Description	The system architecture shall support horizontal scaling by adding server instances to handle increased user load without redesign.
Rationale	Tourist platform traffic is seasonal and can spike significantly during holidays.
Metric / Test	Load balancer and stateless architecture verified; scale-out test with 2× instance doubling
Priority	P1 – Critical

ID	NFR-SC-02
Title	Database Scalability
Description	The database layer shall support scaling to handle a minimum of 1 million records per entity (bookings, users, destinations) without performance degradation.
Rationale	Long-term growth requires the database to handle increasing data volumes efficiently.
Metric / Test	Query performance benchmarked with 1M synthetic records; response within SLA
Priority	P2 – High

ID	NFR-SC-03
Title	Storage Scalability
Description	The system shall support expandable file storage for images and documents without application-level changes.
Rationale	Hotel and destination image uploads will grow over time and require flexible storage.
Metric / Test	Cloud object storage (e.g., S3-compatible) configured; storage expansion tested without downtime
Priority	P2 – High

5.6 Maintainability

ID	NFR-MA-01
Title	Code Modularity
Description	The system codebase shall follow a modular architecture (e.g., MVC or microservices) to enable independent updating of components.
Rationale	Modular code reduces the risk and effort of future changes and bug fixes.
Metric / Test	Architecture review confirms module separation; no cross-module hard dependencies
Priority	P2 – High

ID	NFR-MA-02
Title	Code Documentation
Description	All public APIs, modules, and complex logic shall be documented using inline comments and API documentation standards.
Rationale	Undocumented code increases onboarding time and introduces maintenance risks.
Metric / Test	Code coverage review; all public functions documented; API docs auto-generated
Priority	P2 – High

ID	NFR-MA-03
Title	Logging & Monitoring
Description	The system shall implement centralized logging of all errors, warnings, and critical events, accessible to administrators.
Rationale	Centralized logs are essential for diagnosing production issues quickly.
Metric / Test	Log aggregation tool configured; all error-level events captured and searchable
Priority	P1 – Critical

ID	NFR-MA-04
Title	Automated Testing Coverage

Description	The system shall maintain a minimum of 80% automated test coverage across unit and integration tests.
Rationale	High test coverage reduces regression risk when deploying new features.
Metric / Test	Test coverage report generated by CI/CD pipeline; coverage ≥ 80% enforced
Priority	P2 – High

ID	NFR-MA-05
Title	Deployment Pipeline
Description	The system shall support automated deployment via a CI/CD pipeline to staging and production environments.
Rationale	Manual deployments are error-prone and slow; automation enables reliable releases.
Metric / Test	CI/CD pipeline configured; zero-downtime deployments verified in staging environment
Priority	P2 – High

6. Alternative Solutions & Feasibility Study

6.1 Alternative Solutions

Three alternative system approaches were considered for implementing the Smart Tourism Platform. Each alternative is evaluated based on architecture, features, advantages, disadvantages, and estimated cost.

6.1.1 Alternative A — Off-the-Shelf Tourism SaaS Platform

Field	Details
Overview	License and configure an existing third-party SaaS tourism platform with minimal customization.
Key Features	• Pre-built booking engine for hotels, transport, and tours • Built-in payment integrations • Vendor-managed hosting, security, and maintenance • Limited UI customization
Advantages	• Fast deployment (4–8 weeks) • No infrastructure or DevOps required • Vendor-managed security compliance • Low initial effort
Disadvantages	• Cannot fully satisfy all functional requirements • Limited customization flexibility • No ownership of source code/data • Vendor lock-in risk • Weak support for Sri Lanka-specific integrations
Timeline	4–8 weeks (basic deployment), 3–4 months (configuration completion)

6.1.2 Alternative B — Custom Monolithic Web Application

Field	Details
Overview	Fully custom monolithic application where all modules (auth, booking, admin, payment) are tightly integrated into a single system.
Key Features	• Single codebase system • Server-side rendering (Laravel / Spring MVC) • Relational database (MySQL/PostgreSQL) • Centralized deployment
Advantages	• Full control over system

	• All requirements can be implemented • Simple initial deployment • Easier for small team development
Disadvantages	• Poor scalability under high traffic • Entire system redeployment required for updates • High maintenance complexity • Tight coupling of modules • Performance bottlenecks over time
Timeline	6–9 months

6.1.3 Alternative C — Custom Layered Web Application

Field	Details
Overview	A modern layered architecture system using React frontend, REST API backend, and PostgreSQL database deployed on cloud infrastructure.
Key Features	• React.js / Vue.js responsive frontend • RESTful API backend • MySQL database • JWT authentication with RBAC • CI/CD pipeline integration • Payment gateway integration (Stripe / PayHere)
Advantages	• Fully satisfies all functional requirements • Highly scalable cloud architecture • Modular and maintainable design • Full ownership of system and data • Supports future enhancements (AI, multilingual support) • Enables parallel development
Disadvantages	• Higher initial development effort • Requires DevOps knowledge • More complex setup than monolithic system
Timeline	7–10 months

6.2 Feasibility Study

The feasibility of each alternative is evaluated under four dimensions using a 5-point scale:
1 = Very Low | 5 = Very High

6.2.1 Technical Feasibility

Criteria	Alt A	Alt B	Alt C
Functional Requirement Coverage	Partial (70%)	Full (100%)	Full (100%)
Non-Functional Requirements	Vendor managed	Manual implementation	Built-in design
Scalability	Limited	Vertical scaling only	Horizontal cloud scaling
Customization	Very limited	Full	Full
Technical Risk	Vendor dependency	Monolithic risk	Low (modular design)
Score	3 / 5	4 / 5	5 / 5

6.2.2 Economic Feasibility

Criteria	Alt A	Alt B	Alt C
Initial Cost	Low	Medium	Medium–High
Long-term Cost	High (recurring)	Low	Low–Moderate
Ownership Value	None	Full	Full
ROI Potential	Low	High	Very High
Score	2 / 5	3 / 5	4 / 5

6.2.3 Operational Feasibility

Criteria	Alt A	Alt B	Alt C
Role-Based Access Control	Limited	Full	Full
Mobile Experience	Vendor dependent	Custom required	Fully responsive
Admin Control	Limited	Full	Full
Training Requirement	Low	Medium	Medium

Notification System	Basic	Custom build	Advanced
Score	3 / 5	3 / 5	5 / 5

6.2.4 Schedule Feasibility

Criteria	Alt A	Alt B	Alt C (Recommended)
Deployment Time	4–8 weeks	6–9 months	7–10 months
Parallel Development	Not applicable	Limited	Fully supported
Testing Effort	Low	High	Moderate
Deployment Automation	Vendor managed	Manual	CI/CD enabled
Score	3 / 5	4 / 5	4 / 5

6.3 Comparative Feasibility Summary

Feasibility Dimension	Alt A	Alt B	Alt C
Technical Feasibility	3 / 5	4 / 5	5 / 5
Economic Feasibility	2 / 5	3 / 5	4 / 5
Operational Feasibility	3 / 5	3 / 5	5 / 5
Schedule Feasibility	3 / 5	4 / 5	4 / 5
Overall Score	11 / 20	14 / 20	18 / 20

6.4 Justified Recommendation

Dimension	Justification
Technical	Alternative C satisfies all functional and non-functional requirements with scalable layered architecture (React + REST API + DB).
Economic	Although initial cost is higher, long-term ownership eliminates subscription dependency and increases ROI.
Operational	Fully supports all user roles including tourists, hotels, transport providers, guides, and admins with complete RBAC.
Schedule	Parallel development and CI/CD reduce delivery risks despite longer timeline.

6.5 Conclusion

Alternative C (Custom Layered Web Application) is the most suitable solution with the highest feasibility score (18/20). It ensures full requirement coverage, scalability, maintainability, and long-term sustainability for the Smart Tourism Platform.